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1. Administrative & Study Identification

Full Study Title: A Phase Ib, Open-label, Ascending Dose Study With Step-up Doses to Assess Safety, Tolerability, and Pharmacokinetics of PIT565 in Participants With Systemic Lupus Erythematosus (SLE) **Short Title/Acronym:** PIT565 Ascending Dose Study in SLE **Protocol Number:** CPIT565B12101 **NCT Number:** NCT06335979 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** PIT565 (anti-CD19 x anti-CD3 x anti-CD2 trispecific antibody) **Phase of Development:** Phase Ib **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations / Medical Monitor

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the safety, tolerability, and pharmacokinetics (PK) of PIT565 in participants with Systemic Lupus Erythematosus (SLE). **Secondary Objectives:** To evaluate the pharmacodynamics (PD) of PIT565, particularly the biological activity and achievement of predefined B-cell depletion in participants.

2.2 Background & Rationale To evaluate the therapeutic potential of PIT565 in patients with active SLE. B-cell depletion is a validated approach in treating SLE. PIT565 is a novel trispecific antibody targeting CD19, CD3, and CD2. Compared to traditional bispecific antibodies, CD2 co-stimulation via PIT565 may overcome T-cell exhaustion, enhancing the depth and duration of B-cell depletion and patient responses. This ascending dose study will help identify a safe candidate dose level that achieves 100% predefined B-cell depletion for further clinical investigation.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Uncontrolled (Single-arm across dose cohorts) **Blinding:** Open-label **Randomization:** Non-randomized (Sequential Assignment)

3.2 Planned Enrollment & Duration Target N: 54 Participants (Up to 8 cohorts. 3 sentinel participants per cohort, with up to 3 optional participants. The optimal candidate dose cohort may be enriched with up to 6 additional participants, totaling up to 12 in that cohort). **Number of Sites:** Multicenter (Europe, Rest of World) **Estimated Study Start:** 08-Oct-2024 **Estimated Primary Completion:** 09-Jul-2027

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of SLE according to the 2019 ACR/EULAR criteria. 2. Documentation of SLE autoantibodies. 3. Active SLE disease, as demonstrated by a SLEDAI-2K ≥ 4 at screening. 4. Failure to respond to standard-of-care medicines for the treatment of SLE as detailed in the protocol. 5. Immunization against pneumococcus, meningococcus, influenza, and COVID-19.

4.2 Exclusion Criteria 1. Severe SLE-related organ damage dysfunction or life-threatening disease at screening. 2. Any acute, severe lupus-related flare during screening that needs immediate treatment such as acute CNS lupus (e.g., psychosis, epilepsy) or catastrophic antiphospholipid syndrome. 3. Presence of severe lupus kidney disease as defined by worsening proteinuria or estimated glomerular filtration rate (eGFR) which in the opinion of the Investigator requires immunosuppressive induction or maintenance treatment at screening.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Step-up ascending doses of PIT565. The decision to escalate the dose from one cohort to the next will be based on safety and PD data. **Route of Administration:** Subcutaneous (s.c.) **Duration of Treatment:** Until primary completion or study termination, with dosing escalations dynamically guided by cohort safety tracking.

5.2 Concomitant & Prohibited Therapy Permitted: Premedication prior to PIT565 administration is required. Standard-of-care SLE background therapies (as defined by previous treatment failure thresholds) may be permitted per protocol. **Prohibited:** Concomitant use of other unapproved investigational agents or high-dose immunosuppressive induction therapies for severe acute lupus kidney or CNS disease.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, SLEDAI-2K scoring, autoantibody tests, medical history
Baseline	Day 0	Premedication, First dose of PIT565 (s.c.), baseline PK/PD blood draw
Cohort Dosing	Varies	Sequential assignment, sentinel dosing, step-up dosing administration
Interim Follow-up	Weekly/Monthly	Safety monitoring, Pharmacokinetics (PK), B-cell depletion tracking
End of Study	Q3 2027	Final safety assessment, Exit Interview, IP Accountability

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Assessment of safety and tolerability (incidence of Adverse Events and Dose-Limiting Toxicities) and Pharmacokinetics (PK) of PIT565. **Measurement Method:** Clinical observation, safety labs, and serial plasma concentration profiling of PIT565.

7.2 Secondary & Exploratory Endpoints Pharmacodynamics (PD): Measurement of biological activity, specifically evaluating the dose levels necessary to induce predefined B-cell depletion in 100% of participants. **Efficacy:** Preliminary disease response evaluated via SLE disease activity measures (e.g., SLEDAI-2K reductions).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous cohort-based safety monitoring. The dose escalation decisions rely strictly on safety tracking among sentinel participants prior to optional participant enrollment. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Internal safety review committee evaluates sentinel data to declare candidate dose safety before cohort enrichment.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving at least one dose); PK/PD Set (participants with evaluable samples). **Sample Size Justification:** $N = 54$ is based on standard Phase I/Ib dose-escalation frameworks (up to 8 cohorts $\times$ 3 to 6 participants + cohort expansion of 6 participants at candidate dose) to ensure adequate safety observation and PK/PD profiling. **Handling of Missing Data:** PK/PD data will be handled per Novartis' standard Phase 1 statistical models (e.g., non-compartmental analysis principles).

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and remote Phase I monitoring for Dose-Limiting Toxicities (DLTs) and strict cohort gating. **Audit Trail:** Validated eCRF locking and data clarification forms adhering to global regulatory guidelines for first-in-human/early-phase trials.

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11. Study Identifiers & Governance

Primary ID: CPIT565B12101 **Registry Number:** NCT06335979 **Sponsor/ Lead Organization:** Novartis Pharmaceuticals **Study Title:** PIT565 SLE Ascending Dose Study **Official Regulatory Title:** A Phase Ib, Open-label, Ascending Dose Study With Step-up Doses to Assess Safety, Tolerability, and Pharmacokinetics of PIT565 in Participants With Systemic Lupus Erythematosus (SLE)

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12. Clinical Framework (SLE Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE) **Diagnosis Threshold:** 2019 ACR/EULAR classification criteria & SLEDAI-2K ≥ 4 **Medical Methodology:** B-cell depletion utilizing a CD19 x CD3 x CD2 trispecific antibody **Optimization Target:** 100% predefined B-cell depletion at a declared safe candidate dose

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13. Study Procedures & Methodology

Management Pathway: Ascending dose with step-up dosing utilizing sentinel participants **Education Protocol (HCP):** Protocol training on s.c. administration and premedication requirements **Education Protocol (Patient):** Informed consent and reporting mechanisms for AEs **Quality Audit Mechanism:** Safety and PD gating review prior to dose escalation

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14. Observation & Follow-Up Schedule

Total Duration: Approximately 33 months (Oct 2024 to Jul 2027) **Onsite Visit Cadence:** Frequent early-phase PK/PD blood draw scheduling surrounding step-up dose administrations. **Remote Monitoring Frequency:** Standard ongoing safety check-ins. **Checklist Review Interval:** Pre-escalation safety review milestones per cohort.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]